

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2017

Programme: BE

Full Marks: 100

Course: Digital Signal Analysis and Processing

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) How aliasing is encountered in the processing of discrete-time signals? What can be done to minimize aliasing effect? Explain with the basic elements of digital signal processing where aliasing can be cancelled. 7
- b) A digital communication link carries binary-coded words representing samples of an input signal
 $x_a(t) = 3\cos 5000\pi t + 2\sin 6000\pi t$
The link is operated at 18,000 bits/s and each input sample is quantized into 512 different voltage levels. 8
- i. What is the discrete-time signal obtained after sampling?
- ii. What is the resolution Δ ? 2
2. a) Prove that multiplication of a sequence $x[n]$ by $e^{j\omega_0 n}$ is equivalent to a frequency translation of the spectrum $X(e^{j\omega})$ by ω_0 . 7
- b) An LTI, causal, second order discrete-time system is described by the difference equation 8
- $$y[n] = 6y[n-1] - 8y[n-2] + 16x[n]$$
- Calculate its step response $s[n]$. 2
3. a) Prove that necessary and sufficient condition for stability is $\sum_{-\infty}^{\infty} h[k] < \infty$. 7
- b) Why is DFT preferred over DTFT in the analysis of discrete - time signals? Also determine the DFT of the signal $x[n] = u[n] - u[n-4]$ using DIF-FFT algorithm. 8
4. a) What is the difference between linear and circular convolution? Find the linear circular convolution of the following sequences: 7
- $$x[n] = \{1, 0, 0, 1\} \text{ and } h[n] = \{2, 0, 2\}$$

Or

Show that multiplication of two discrete time signals results in circular convolution of their DFTs

- 5 b) Draw the lattice-ladder structure for the following IIR system.

$$H(z) = \frac{1 - 0.8z^{-1} - 0.15z^{-2}}{1 + 0.1z^{-1} - 0.72z^{-2}}$$

Is the system stable?

- 7 5. a) Design a digital Butterworth low pass filter satisfying the constraints
- $$\begin{cases} -3.011\text{dB} \leq |H(e^{j\omega})| \leq 0\text{dB} & 0 \leq \omega \leq \frac{\pi}{2} \\ |H(e^{j\omega})| \leq 10.45\text{dB} & \frac{3\pi}{4} \leq \omega \leq \pi \end{cases}$$

With $T = 1\text{sec}$ using bilinear transformation method. Realize the filter using the most convenient realization form.

- 7 b) Given an analog system

$$H_a(s) = \frac{s^2 + 1.4s + 9.6}{(s + 0.5)(s^2 + 0.4s + 9.4)}$$

Obtain a digital filter by using impulse invariant method. Assume $T = 1$.

- 6.4 6. a) Explain briefly the design of linear phase FIR filter by frequency Sampling method with proper example
- 6.3 b) Design a linear FIR filter using Kaiser window to meet the following specifications:

$$\begin{aligned} 0.98 \leq |H(e^{j\omega})| \leq 1.02, & \quad \text{for } 0 \leq |\omega| \leq 0.2\pi \\ |H(e^{j\omega})| \leq 0.03, & \quad \text{for } 0.3\pi \leq |\omega| \leq \pi \end{aligned}$$

7. Write short notes on: (Any two)

- a) Properties of LTI system. 2
- b) Computational complexity of DFT. 4
- c) Limit Cycle Oscillation Effect. 7